Week 5: Topic Models and Mixed-Membership Models
Jaisel Singh (js6897)
October 14, 2026

What stood out to me in Blei's review was the shift from clustering a document to clustering its words. A topic is a distribution over a fixed vocabulary, every document in the collection draws on the same topics, and what distinguishes one document from another is the proportion in which it uses them. Because a topic is chosen separately for each word, a single document can express several topics at once. That is what mixed membership means, and it is what a model assigning one cluster per document cannot represent. The generative story is short: draw topic proportions for a document, then for each word draw a topic assignment and draw a word from the corresponding topic. I noticed the Dirichlet-categorical structure sitting underneath both of those draws.

What I found most useful was how cleanly the model separates from the computation. The words are observed and the topic structure (the topics, the per-document proportions, and the per-word assignments) is hidden, so fitting the model means computing a posterior over that hidden structure. Blei describes this as reversing the generative process. The obstacle is the evidence in the denominator, which sums the joint distribution over every way of assigning each observed word to a topic, and that count grows exponentially with the collection. This is the same wall we hit last week, and the responses are the same two families: sampling, usually Gibbs sampling, or variational approximation. It makes the MCMC material feel less like a separate topic and more like the machinery these models need in order to be fit at all.

Boyd-Graber, Hu, and Mimno's survey complicated my sense of what a good fit means here. They report that held-out likelihood can correlate negatively with how interpretable readers find a model, since a model with more topics can predict held-out words better while being harder for a person to read. Automated coherence scores stand in for human judgment, but they do not establish that a model suits a particular application. My remaining question is how to set the number of topics and the Dirichlet hyperparameters when the quantity I can compute is not the quantity I actually care about.

Readings: Blei (2012), *Probabilistic Topic Models*; Boyd-Graber et al. (2017), *Applications of Topic Models*.